English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics — Written exam, 01.02.2023

The exact grading formula depends on the results. The minimum score required to pass is 1 point. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

Integration formulas: $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$, $\mathrm{d}f(x) = f'(x)\mathrm{d}x$.

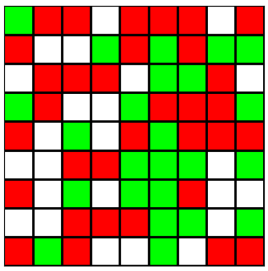
Problem 1. A standard roulette wheel has 18 red, 18 black and one green number. For a successful bet on black/red, the casino pays twice the stake.

1. (0.5 points) Assume all bets are BGN 1. How many bets per day must the casino accept so that its profit is at least BGN 1000 with probability 90%? With this many bets, what is the probability that the casino makes a loss on a given day?
2. (0.25 points) One may also bet on a single number, with a payout of 36 times the stake. Which would you prefer: betting on one number or on black/red? (Compare the expectations and variances and make your choice using them.)
3. (0.25 points) If you have BGN 5, which would you prefer: five bets on a number over five spins, bets on five different numbers in one spin, or a BGN 5 bet on black/red in one spin?

Problem 2. Let the joint density of X and Y be $f_{X,Y}(x,y) = cx^3y$ for $x, y \geq 0$, $x + y \leq 1$, and zero elsewhere, where c is a constant. Find:

1. (0.5 points) c, the density of X and the expectation of Y. Explain how you would calculate $Cor(X,Y)$;
2. (0.25 points) $\mathbb{E}(Y|X = 1/2)$ and $\mathbb{E}(X|X = 1/2)$;
3. (0.25 points) the density of the random variable $Z = XY$.

Problem 3. The cells of a 9×9 grid are coloured at random in one of the colours white, green and red.



1. (0.5 points) What is the expected number of cells coloured white or green? What is the probability that there are at least 60 such cells?

   A horizontal (vertical) flag is three consecutive horizontal (vertical) cells coloured white, green and red from left to right (top to bottom).
2. (0.25 points) Is the number of horizontal flags binomially distributed? If so, what are its parameters?
3. (0.25 points) What is the expected number of horizontal flags?
4. (0.25 points) Are the numbers of horizontal and vertical flags independent? What is the expected total number of horizontal AND vertical flags?

Problem 4. Let $X_1, X_2, \ldots$ be independent $Exp(1)$ random variables and $S_n = X_1 + \cdots + X_n$.

1. (0.5 points) Find the density of S_2 and prove by induction that the density of S_n is
$$f_{S_n}(x) = \frac{x^{n-1}}{(n-1)!} e^{-x} \mathbb{1}_{\{x>0\}}.$$
2. (0.25 points) Let $N_t := \max\{n : S_n \leq t\}$. Prove that $\mathbb{P}(N_t \geq n) = \mathbb{P}(S_n \leq t)$ and that $\mathbb{P}(N_t = n) = \mathbb{P}(S_n \leq t) - \mathbb{P}(S_{n+1} \leq t)$.
3. (0.25 points) Find the distribution of N_t.